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| **Teacher:** | Ray Joshua Balingkit | **Subject:** | Design & Technology |
| **Week:** | 4 (Term 1) | **Year:** | 10–11 |
| **Link:** | | **Cover:** | |

## LESSON OVERVIEW

### What should students know/do by the end of this week's lessons?

Students will distinguish energy generation and storage and explain examples used in products. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will compare energy options against the needs of a defined product. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

### End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

### LESSON SLOTS - TUESDAY AND WEDNESDAY

| Lesson | Date | Learning Objective | The Lesson Flow | Differentiation |
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| 1 | 15 September 2026 | By the end of this lesson, students will be able to distinguish energy generation and storage and explain examples used in products. | **H:** Show pictures of a solar panel, battery and electric motor. Ask which converts incoming energy, which stores it and which produces movement.<br><br>**ID:** Use the book's energy section to distinguish renewable and finite energy resources and generation from storage. Explain that energy is transferred or converted rather than created. A battery stores energy chemically; a solar cell converts light to electrical energy, and a motor converts electrical energy into movement with losses.<br><br>**WD:** Trace a solar garden light from sunlight through its panel, rechargeable battery and lamp. Model a simple energy-flow diagram and discuss what happens at night or after several cloudy days. Distinguish the panel's | **Support:** Provide labelled component cards and arrows. Complete the torch example together and offer a generation-storage-output word bank for the independent diagram.<br><br>**Extension:** Students explain how low output, charging opportunities and product operating time influence the suitability of an energy arrangement without assuming a larger battery is always better. |

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| | | | role from the battery's role.<br><br>**YD:** Students classify supplied energy-source and storage examples, then draw energy-flow diagrams for a torch and a solar light. They annotate one limitation of each arrangement and explain why a storage component may be useful when the source is intermittent.<br><br>**C:** Ask students why adding a battery does not generate a new energy supply. Check that diagrams show sensible transfers and that renewable does not mean free of manufacturing or disposal impacts. | |
| 2 | 16 Septemb er 2026 | By the end of this lesson, students will be able to compare energy options against the needs of a defined product. | **H:** Present a portable reading light used where charging is sometimes unavailable. Ask what the designer needs to know before choosing its energy arrangement.<br><br>**ID:** Explain selection criteria including portability, operating time, charging access, maintenance, purchase cost and end-of-life handling. Distinguish a primary cell from a rechargeable cell and discuss using the specified charging system. Classroom comparisons use case information; no cells are opened or improvised charging attempted.<br><br>**WD:** Compare a replaceable-cell light, USB-rechargeable light and solar-rechargeable light against a supplied user scenario. Model a justified choice using operating conditions and acknowledge the weakness of the chosen option in a different setting. | **Support:** Use a matrix with clear criteria and two options initially. Highlight relevant case facts and provide a recommendation frame linking choice, user condition and limitation.<br><br>**Extension:** Students suggest a fair test of operating time, controlling brightness and starting condition, and explain how measured results could change their recommendation. |

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| | | | **YD:** Students complete a comparison matrix using teacher-prepared product data. They recommend an arrangement for a reading light, justify it against three needs and identify one condition that would require a different solution. Partners check whether each claim is supported by the data provided.<br><br>**C:** Collect the recommendation and ask which user requirement mattered most. Check that students distinguish energy storage capacity from generation and recognise the limits of a comparison without complete performance data. | |
| **3** | | | | Support:<br><br>Extension: |
| **4** | | | | Support:<br><br>Extension: |

## MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.3 How energy is generated and stored
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



## END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?